AI COURSEWORK

Design and Development of Specialized Artificial Neural Networks for Industry-Specific Applications

Assessment weighting: 100% of the module mark | Individual coursework

1. Purpose of the Coursework

This coursework is the sole summative assessment for the module. It is designed to assess your ability to take an industry-relevant problem from problem formulation through data preparation, neural-network design, experimentation, evaluation, deployment, explainability, and consideration of modern generative/foundation models.

The emphasis is not on achieving the highest possible predictive score. Marks are awarded for sound technical reasoning, fair experimentation, appropriate model selection, evidence-based conclusions, reproducibility, and your ability to defend your decisions.

Read the complete brief before selecting a problem. Your problem and dataset must allow you to answer all six questions. You must obtain lecturer approval for your proposed problem and dataset before beginning the main implementation.

2. Project Requirements at a Glance

1. Choose one specific real-world application and identify the intended user and the decision supported by the system.
2. Collect or obtain a dataset that is appropriate for the chosen task.
3. Perform data preparation, EDA, and explicit leakage checks.
4. Design and train an ANN specifically for the chosen problem.
5. Train and tune at least one classical machine-learning baseline using a comparable level of tuning effort.
6. Conduct controlled experiments, including at least four unsuccessful attempts, and maintain an experiment log.
7. Evaluate models using a clearly justified primary metric and report the final evaluation on a single held-out test set.
8. Deploy the ANN in a functional web application that produces predictions at request time.
9. Implement an appropriate explainable-AI method in the application and validate that the explanations are meaningful.
10. Investigate an appropriate generative model and compare it fairly with your own model or use it for an approved adjacent task where direct application is not meaningful.
11. Submit reproducible code, an experiment log, the dataset or reproducible download procedure, a report, and an 8-minute maximum demonstration video, and defend the work in a live presentation.

3. Choosing Your Problem

Select a specific application that requires the development of a specialized artificial neural network (ANN) system. Start with a problem, not with a dataset. Before you commit to a project, you must be able to complete both statements below with specific answers:

- *"The user of this system is _____."*
    - The user must be a specific role, such as a radiologist, claims officer, maintenance engineer, teacher, or supply-chain planner. "The public", "customers", or "companies" is not sufficiently specific.
- *"When the model produces a prediction, that user does _____ differently as a result.*
    - Describe the actual decision or action that changes because of the prediction.

If you cannot complete both statements concretely, the proposed project is an exercise rather than an application and will not provide a sufficient basis for Questions 1, 4, and 5.

Possible application areas include:

- **Medical AI:** diagnostics from images or signals, predictive analytics, personalized medicine.
- **Educational AI:** intelligent tutoring, automated assessment of free-text answers, personalized learning.
- **Legal AI**: document analysis, clause extraction, case-related prediction, contract management.
- **Mathematical AI**: symbolic reasoning, theorem proving, numerical analysis.
- **Enterprise AI**: CRM, supply-chain optimization, financial forecasting, demand prediction.
- **Industrial and other domains**: predictive maintenance, quality inspection, agriculture, energy, telecoms.

3.1 Preferred Data Modalities

You are strongly encouraged to select a problem involving unstructured or genuinely sequential data: images, video, free text, audio, sensor readings, logs, or time series. Such data provides a natural opportunity **to apply neural architectures covered in the module**.

Tabular data is permitted only when there is a substantive reason for a neural approach.

3.2 Datasets You May Not Use

Do not use datasets for which complete end-to-end tutorials already exist and which would substantially answer the coursework for you.

The following are explicitly excluded: *Titanic; Iris; Wine; Boston Housing; California Housing; MNIST; Fashion-MNIST; unmodified CIFAR-10; IMDB sentiment; Breast Cancer Wisconsin; Pima Diabetes; UCI Heart Disease; Dogs vs. Cats*.

If you are uncertain whether a dataset is acceptable, obtain written approval from the lecturer before proceeding.

3.3 Approval and Uniqueness

- **Submit your proposed problem**, intended user, decision supported, dataset, target variable/task, and proposed data modality to the lecturer for approval before substantial implementation.
- You are responsible for confirming that all six assessed questions can be answered for your proposed problem, including Question 6.
- Changing the approved problem or dataset later requires lecturer approval.

4. Dataset Collection and Preparation

- Collect, create, or obtain a dataset appropriate to the approved problem.
- Personal, confidential, or health-related data must be appropriately anonymised before submission or use.
- Document the dataset source, license/permission, number of samples, features/modalities, target definition, class distribution where relevant, and important data-quality limitations.
- Perform preprocessing appropriate to the modality and document every transformation that affects model input.
- Check explicitly for data leakage, including features unavailable at prediction time and duplicate or near-duplicate samples crossing train/validation/test boundaries.
- Where observations belong to individuals, devices, sites, patients, customers, documents, or other groups, prevent related observations from leaking across splits.

5. Model Development

Develop an ANN that is appropriate for the structure and modality of your approved problem.

- Explain why the selected architecture is appropriate rather than simply naming the architecture.
- Train at least one classical machine-learning baseline.
- The ANN and classical baseline must be evaluated on the same held-out test set and using the same primary metric.
- Transfer learning and fine-tuning of pretrained neural backbones are encouraged where technically appropriate. There are no marks for training from scratch when fine-tuning is the sensible approach.

6. Project Rules: Experimental Integrity and Reproducibility

- **Single split:** Define one train/validation/test split and use that same split for the classical baseline, all neural-network variants, and the GenAI evaluation.

- **Test set isolation:** The test set may be used only for the final evaluation. All model selection, hyperparameter decisions, prompt selection, threshold selection, and other intermediate decisions must be made using training/validation data only.
- **Seeds and repeated runs:** Fix and report random seeds. If results vary materially by seed, report the mean and spread over at least three runs.
- **Comparable tuning effort**: Do not compare a heavily tuned ANN with an untuned classical baseline. State the approximate number of configurations evaluated and the search space/budget for each model family; the effort must be of the same order of magnitude.
- **Honest reporting**: If the classical baseline outperforms the ANN, report this without modification and explain the result using evidence from the data and experiments. A fair classical model beating the ANN can receive full marks.
- **Experiment log**: Record every substantive training/evaluation run, including unsuccessful runs. Each entry must state what was changed, the relevant hyperparameters, the validation result, and what was learned.
- **Reproducibility:** Your submitted code must contain enough information for another person to reproduce the reported results, subject to reasonable limitations such as unavailable proprietary data or paid external APIs.

7. Use of Generative AI Tools

Generative AI tools may be used as productivity or development aids, but their use must be declared in an appendix. You remain responsible for every item of code, result, interpretation, citation, and claim in your submission and must be able to explain the submitted work during the presentation.

- Examples that must be declared include using ChatGPT, Claude, Gemini, Copilot, or similar tools for coding, debugging, analysis, writing, or brainstorming.
- Using a foundation/generative model as the assessed model in Question 6 is part of the coursework itself and must be documented as such.
- Generated material must not be submitted without understanding and verifying it.

8. Deliverables

- **Code:** All scripts/notebooks used for preprocessing, training, evaluation, and deployment, with documentation, a README, and a requirements/environment file.
- **Experiment log:** A complete table of substantive experiments, including failed experiments. Any result discussed in the report must be traceable to the log.
- **Dataset:** The dataset used. If the dataset exceeds 500 MB, submit a download script and a sufficiently representative sample that allows the code to be tested.
- **Demonstration video**: A screen recording of the working system, maximum 8 minutes. It must show a real-time prediction, the explanation feature, and at least one incorrect prediction.
- **Report**: Maximum 25 pages excluding appendices. Structure the report in the order of Questions 1–6 so each assessed component can be located easily.

- **Live presentation**: Present the project and be prepared to answer questions about your data, architecture, experiments, code, results, deployment, XAI, and GenAI decisions.

9. Assessment

Question 1 – Dataset and Exploratory Data Analysis (10 marks)

- Describe the dataset, including its source, licence/permission, structure, size, target, and relevant characteristics.
- Perform a comprehensive EDA appropriate to the modality.
- Identify the relative consequences/costs of false positives and false negatives where classification or detection is involved; for other tasks, identify the relevant decision consequences.
- Define and justify the primary evaluation metric(s). Explain why the chosen metric reflects the user's decision and data characteristics.
- Report your leakage checks and how data-quality problems were handled.
- End the EDA with three findings that materially constrain your modelling choices. Descriptive plots or statistics without modelling consequences will receive limited credit.

Question 2 – Solution Design (20 marks)

2.a Design and Architect a Suitable ANN (10 marks)

- Present the architecture, including layers, dimensions, activations, parameter count, and a clear architecture diagram.
- Explain what assumptions the architecture makes about the data and why those assumptions suit the problem.
- Justify the major design decisions individually, including depth, width, normalisation, regularisation where used, output layer, and loss function.
- Ensure the loss/output formulation is consistent with the task and the decision consequences identified in Question 1.

2.b Discuss Alternative Approaches (10 marks)

- Discuss meaningful alternative approaches considered and explain why the selected approach was chosen.
- Justify why an ANN is appropriate based on the modality, volume, structure, and task.
- Before evaluating the final results, state which model family (ANN or classical baseline) you expect to perform better on the primary metric and explain why. Marks depend on the quality of the reasoning, not on whether the prediction is correct.
- For each important rejected alternative, state what characteristics would have made it appropriate and what property of your problem made it less suitable.

Question 3 – Model Development, Evaluation and Optimization (20 marks)

3.a Development, Baseline Comparison and Error Analysis (10 marks)

- Describe the initial neural model and how it was improved.
- Present and interpret learning curves. Use evidence to determine whether the model is underfitting, overfitting, or otherwise limited.
- Present the classical baseline, including its tuning search space and number of configurations.
- Provide a comparison table containing: an appropriate trivial/reference benchmark, the classical baseline, the initial ANN, and the final ANN. Report the primary metric, training time, and inference latency.
- Conduct error analysis on the final model. Characterise errors by relevant class, subgroup, input type, data-quality issue, or other meaningful factor.

3.b In-depth Study of Optimization Techniques (10 marks)

- Discuss the strengths and weaknesses of relevant optimization techniques and compare them using evidence from your own experiments.
- Investigate the effects of at least three hyperparameters chosen from learning rate, batch size, network depth, or other lecturer-approved parameters. Do not merely describe them; support conclusions with your results.
- Compare at least two optimisers empirically and relate observed behaviour to their underlying mechanics.
- Report at least two substantive approaches that did not work and diagnose why. Failed experiments receive credit when they are well designed and correctly analysed.
- Keep the experiment log complete and consistent with the results reported in the final report.

Question 4 – Web Application Implementation (10 marks)

- Implement a functional web application with a frontend and backend that interacts with the trained ANN.
- The application must allow the intended user identified in Question 1 to submit an input and receive a prediction.
- The prediction must be generated by the trained model at request time. Hard-coded or precomputed predictions are not acceptable.
- Display an appropriate prediction score/probability or uncertainty measure and explain what it represents. Do not describe an uncalibrated probability as a guaranteed confidence level.
- Clearly state the model's relevant limitations to the user.
- Test normal inputs, invalid inputs, and inputs that are materially outside the training distribution where such testing is technically meaningful.

Question 5 – Explainable AI (20 marks)

5.a Discussion of XAI Methodologies (5 marks)

- Compare at least three XAI approaches, including both model-agnostic and architecture-specific methods where meaningful for your problem.
- For each approach, discuss what it explains, computational cost, and important limitations.
- Justify the selected method with reference to your architecture and intended user. The explanation must be appropriate for the user's level of technical expertise and decision context.

5.b Implementation and Validation of XAI (15 marks)

- Implement an explanation feature for the ANN and make it accessible through the web application.
- Select a method appropriate to the architecture and data modality, such as class-activation/saliency methods for convolutional models, attribution methods for sequence models, or suitable attribution methods for dense models.
- Sanity-check the explanations using at least two cases. For example, perturb or occlude the highlighted region/features and determine whether the model's prediction changes in the expected direction.
- Include at least one incorrect prediction and analyse what the explanation suggests about the model's behaviour.
- Present the explanation in an intuitive UI/UX that is appropriate for the intended user.
- Do not claim that an explanation proves causal reasoning or that it necessarily represents the true reason a prediction occurred; discuss the limitations of the selected method.

Question 6 – Generative AI Models (20 marks)

6.a Apply a Generative Model (10 marks)

- Apply a suitable pretrained generative or foundation model to the approved problem where a technically meaningful application exists. Depending on the modality, this may be an LLM, vision-language model, speech model, or another appropriate foundation model.
- Use the same underlying held-out test examples and the same primary evaluation metric as the ANN wherever a technically meaningful comparison is possible.
- If prompting is involved, document the prompts and select/finalize them using validation data only. Do not use test results to iteratively design prompts.
- If no generative model can be applied meaningfully to the main task, you may instead use one for an approved adjacent task, such as synthetic-data generation, labelling/label auditing, feature extraction, or generation of explanations.
- Document the model/version, relevant configuration, prompt(s) where applicable, external services used, and any important limitations or costs.

6.b Comparative Analysis and Innovation Opportunities (10 marks)

- Compare the GenAI model approach and your ANN using evidence. Where applicable, compare predictive performance, inference latency, cost per prediction, training-data requirements, privacy implications, and dependence on third-party services.
- Do not assume that the GenAI model approach must outperform the ANN, or vice versa. Your conclusion must follow from the evidence.
- Identify the conditions under which each approach would be appropriate for deployment.
- Identify at least one realistic opportunity where a GenAI/foundation model could extend your system beyond the capabilities of your own ANN.

10. Mark Distribution

Question	Component	Marks
Q1	Dataset and EDA	10
Q2.a	ANN design and architecture	10
Q2.b	Alternative approaches and justification	10
Q3.a	Model development, baseline comparison and error analysis	10
Q3.b	Optimisation study	10
Q4	Web application implementation	10
Q5.a	XAI methodologies	5
Q5.b	XAI implementation and validation	15
Q6.a	Generative/foundation model implementation	10
Q6.b	Comparative analysis and innovation opportunities	10
	TOTAL	100

11. Presentation and Individual Verification

The live presentation is part of the assessment process. You must be able to explain and defend your submitted work.

- You may be asked to explain any design decision, experiment, result, code component, deployment decision, or XAI/GenAI choice.
- You may be asked to demonstrate the application and explain an individual prediction.
- The lecturer may ask you to make or explain a small change to the system or code to verify your understanding.

- You are responsible for all submitted material, including material produced with AI-assisted tools.

12. Academic and Technical Integrity

- Do not manipulate data splits, metrics, baselines, experiments, or results to create a more favorable conclusion.
- Do not weaken a baseline or selectively omit unsuccessful experiments.
- Do not use data that you are not authorized to use.
- Do not present generated code, explanations, or written material as your own understanding without verification and declaration of AI-tool use.
- Any result presented in the report must be reproducible from the submitted code and experiment record, subject to documented external-data or service limitations.

13. Final Student Checklist

☐ My problem has an identified user and a clearly stated decision/action supported by the prediction.
☐ My dataset and task were approved by the lecturer.
☐ I used one fixed train/validation/test split throughout the project.
☐ I prevented leakage and documented my checks.
☐ I justified my primary evaluation metric.
☐ I trained both an ANN and a classical baseline with comparable tuning effort.
☐ I recorded all substantive experiments, including failed experiments.
☐ I evaluated the final models only once on the held-out test set.
☐ I conducted error analysis.
☐ My web application generates predictions at request time.
☐ My XAI method is appropriate to my model and has been sanity-checked.
☐ I included at least three incorrect predictions and analyzed it with XAI.
☐ I completed Question 6 using a meaningful GenAI application or obtained prior approval for an adjacent task.
☐ My code, environment information, README, dataset/license information, and experiment log are included.
☐ My demonstration video is no longer than 8 minutes and shows the required features.
☐ My report is no longer than 25 pages excluding appendices and follows Questions 1–6.
☐ I have declared my use of generative AI tools.
☐ I can explain and defend every part of my submission.

14. Closing Principle

This coursework assesses judgement as well as implementation. A successful project is not defined by obtaining the highest accuracy or by proving that a neural network is always better than a classical model. You are expected to frame a meaningful problem, understand

the data, choose an architecture for technically defensible reasons, conduct fair experiments, analyse failures, deploy the system, examine its explanations, and evaluate whether modern foundation models add value. Your conclusions should be determined by evidence.